

MF RECOGNITION IS Π_2^0 -COMPLETE

SAUERS

A countable group is MF if it has finite-dimensional unitary models that are asymptotically multiplicative and separate nonidentity elements in operator norm. Deciding whether a finite presentation defines an MF group is Π_2^0 -complete: it is impossible to construct an algorithm that decides from a finite presentation whether the group it defines is MF, or even one that enumerates the presentations of MF groups or those of non-MF groups. A presentation defines an MF group if and only if every finite word challenge has a word-problem witness or a Gaussian-rational unitary model with decidable norm certificates. In the other direction, a computable family of finite presentations $\hat{R}_e$ defines MF groups if and only if the e -th program halts on infinitely many inputs: for finitely many inputs the group contains the finitely presented non-MF group of [NMF, Theorem C], and for infinitely many inputs it is MF by a permanence theorem for HNN extensions whose edge isomorphism is implemented by a unitary of a norm matrix corona.

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1. INTRODUCTION

Not every countable group is MF [NMF]: the group $H = \text{EL}_{12}(L_{\mathbb{F}_2}(1, 2))$ is finitely generated and simple, has property (T), and every homomorphism from H to an MF group is trivial, and there is a finitely presented torsion-free group E with the same last property [NMF, Theorem C]. This paper determines the complexity of recognizing MF groups from finite presentations.

Theorem A (MF recognition is Π_2^0 -complete). *Deciding whether a finite presentation defines an MF group is Π_2^0 -complete, and the complementary problem is Σ_2^0 -complete. In particular, it is impossible to construct an algorithm that decides from a finite presentation whether the group it defines is MF, or even one that enumerates the presentations of MF groups or those of non-MF groups. There is a computable map $e \mapsto \hat{R}_e$ such that $\hat{R}_e$ presents an MF group if and only if the e -th partial computable function has infinite domain.*

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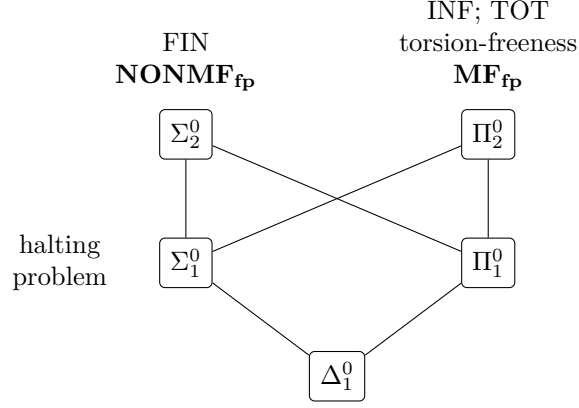


Figure 1: The arithmetical hierarchy. Lines indicate inclusions, and each listed problem is complete at its level. The two sets of Theorem A appear in bold: $\text{MF}_{\text{fp}} \in \Pi_2^0$ by the finite certificates of Section 3, and the family $\hat{R}_e$ of Section 5 reduces INF to MF_{fp} and FIN to NONMF_{fp} .

A finite challenge is a pair (W, k) , where W is a finite list of words and k is a precision index. An answer supplies a word of W together with a finite word-problem witness that it is trivial, or Gaussian-rational unitary matrices for the generators together with finite certificates that every relator has norm defect less than $1/(k+1)$ and every word of W has norm displacement greater than $1/3$. All checks are primitive recursive, and a finite presentation defines an MF group if and only if every coded challenge has an accepted coded answer. So $\text{MF}_{\text{fp}} \in \Pi_2^0$ and $\text{NONMF}_{\text{fp}} \in \Sigma_2^0$ (Section 3).

The finite presentation $\hat{R}_e$ of Section 5 is written on generators and relators of a direct product $\Gamma_e \times F$, with F free of rank three, and the group it presents is an HNN extension of $\Gamma_e \times Q_e$, where Q_e is a three-generator recursively presented quotient of F and Γ_e is a central HNN extension of a product of free groups over a subgroup built from a Mihailova subgroup, which detects the word problem of Q_e ; and Q_e is residually finite or contains E according to whether the e -th domain is infinite or finite. If the domain is finite, then $\hat{R}_e$ contains E and is not MF. If the domain is infinite, the edge isomorphism of the HNN extension is implemented by a unitary of a norm matrix corona, and $\hat{R}_e$ is MF by Theorem 4.2.

Relation to prior work. Decision problems for finitely presented groups go back to Dehn, and the theorem of Adian and Rabin [Ad, Ra] shows that no Markov property (one that some finitely presented group has and that no group containing some fixed finitely presented group has) can be recognized from a finite presentation. Once a finitely presented non-MF group exists, the MF property is Markov and so undecidable; Theorem A determines the exact level. A classical Π_2^0 -complete recognition problem for finitely presented

groups is torsion-freeness, by Lempp [Lem]. Korchagin proved that the MF property is closed under direct limits [Kor, Proposition 13], so every finitely generated non-MF group is a quotient of a finitely presented one; the seed E used below is the explicit finitely presented group of [NMF, Theorem C]. The operator algebra enters in the passage from recursive to finite presentations: the HNN extensions of Section 5 are chosen so that Theorem 4.2 applies to them.

The certificate checker of Section 3 and the constructions of Section 5 are formalized in Lean 4 [Lean]. Proofs whose remaining verification is routine are given in outline and refer there.

2. MF GROUPS

A countable group G is *MF* if there are positive integers d_n and maps $V_n: G \rightarrow \mathcal{U}(d_n)$, with $V_n(1) = 1$, such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \rightarrow 0 \quad (g, h \in G),$$

and

$$\limsup_n \|V_n(g) - 1\| > 0 \quad (g \in G \setminus \{1\}).$$

This is the operator norm notion of MF group introduced by Carrión–Dadarlat–Eckhardt [CDE], and MF group always means this notion below. The strong convergence convention also requires the matrix models to recover the norms of the left regular representation [GKM, Sch], so a group that is not MF in the present sense is not MF in that sense either. Equivalently, G embeds in the unitary group of the norm matrix corona associated to the sequence $\mathbf{d} = (d_n)$,

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}),$$

where $\prod_n M_{d_n}(\mathbb{C})$ is the C^* -algebra of bounded sequences and $\bigoplus_n M_{d_n}(\mathbb{C})$ is its ideal of sequences with $\|x_n\| \rightarrow 0$. A separable C^* -algebra is called MF if it embeds as a C^* -subalgebra of a norm matrix corona [BK]. Restricting models to a subgroup shows that subgroups of MF groups are MF.

Throughout, the binary Leavitt algebra $L_{\mathbb{F}_2}(1, 2)$ is the unital $\mathbb{F}_2$ -algebra generated by s_0, s_1, t_0, t_1 subject to $t_i s_j = \delta_{ij}$ and $s_0 t_0 + s_1 t_1 = 1$, and $H = \text{EL}_{12}(L_{\mathbb{F}_2}(1, 2))$ is the subgroup of $\text{GL}_{12}(L_{\mathbb{F}_2}(1, 2))$ generated by the elementary matrices $e_{ij}(a) = 1 + aE_{ij}$, $i \neq j$. The idempotent $1 - s_0 t_0 = s_1 t_1$ is full, since $t_1 s_1 t_1 s_1 = 1$, so by [NMF, Theorem 3.4], applied with $s = s_0$ and $t = t_0$, every homomorphism from H to an MF group is trivial; in particular H is not MF.

3. LOCALITY AND FINITE CERTIFICATES

The MF property is defined through sequences of models on all of G , and it is determined by finite data with a fixed separation constant. The following formulation is derived from Korchagin [Kor, Propositions 5 and 7].

Lemma 3.1 (locality with fixed separation). *A countable group G is MF if and only if for every finite set $F \subseteq G$ containing 1 and every $\varepsilon > 0$ there are $d \geq 1$ and a map $V: F \rightarrow \mathcal{U}(d)$ with $V(1) = 1$ such that*

$$\begin{aligned} \|V(gh) - V(g)V(h)\| &\leq \varepsilon \quad \text{whenever } g, h, gh \in F, \\ \|V(g) - 1\| &\geq 1 \quad \text{for } g \in F \setminus \{1\}. \end{aligned}$$

Proof. Maps for an exhaustion $(F_n, 1/n)$ of G , extended by 1 outside F_n , form an MF model. Conversely, Korchagin's amplification [Kor, Proposition 5] first replaces V_n by $V_n \oplus 1$, so that 1 lies in the spectrum of each $V_n(g) \oplus 1$ and the diameter of that spectrum is at least $\|V_n(g) - 1\|$, and then passes to an iterated adjoint representation $\text{Ad}^k \circ (V_n \oplus 1)$. The summand 1 is needed because Ad of a scalar unitary is the identity. For a fixed nonidentity element g and a subsequence on which $\|V_n(g) - 1\|$ is bounded below, a number k of iterations depending only on that lower bound raises the separation of g to $\sqrt{2}$, while each iteration multiplies the defects by a constant. A block direct sum of late amplified models, one summand for each $g \in F \setminus \{1\}$, gives the stated map [Kor, Proposition 7]. $\square$

We use the Kleene–Mostowski arithmetical hierarchy in its standard form [So]: a set $A \subseteq \mathbb{N}$ is Σ_1^0 if $A = \{x : \exists y R(x, y)\}$ for a decidable relation R , it is Π_1^0 if its complement is Σ_1^0 , and Σ_2^0 and Π_2^0 are the classes $\{x : \exists y \forall z R(x, y, z)\}$ and $\{x : \forall y \exists z R(x, y, z)\}$ with R decidable. Fix an effective coding of finite presentations by natural numbers, under which a code determines computably its number of generators and its finite list of relator words, and write G_P for the group presented by the code P . Let MF_{fp} be the set of codes P such that G_P is MF, and NONMF_{fp} its complement. Korchagin's Proposition 8 [Kor] is the finite-presentation form of Lemma 3.1; the certificates below add normal-closure witnesses for the trivial words, and the relation becomes decidable.

Proposition 3.2 (finite certificates). *There is a decidable relation $C(P, n, c)$ on triples of natural numbers such that a code P lies in MF_{fp} if and only if for every n there is c with $C(P, n, c)$. Consequently $\text{MF}_{\text{fp}} \in \Pi_2^0$ and $\text{NONMF}_{\text{fp}} \in \Sigma_2^0$.*

Proof. Let $P = \langle x_1, \dots, x_s \mid r_1, \dots, r_m \rangle$. A *challenge* is a pair (W, k) , where W is a finite list of words in the generators and $k \in \mathbb{N}$. The challenge is answered if either

- (1) some $w \in W$ represents 1 in G_P , or
- (2) there are unitary matrices $U_1, \dots, U_s$ such that

$$\|r_i(U) - 1\| < \frac{1}{k+1} \quad (1 \leq i \leq m), \quad \|w(U) - 1\| > \frac{1}{3} \quad (w \in W).$$

Then

- (1) G_P is MF $\iff$ every finite challenge (W, k) is answered.

For the forward implication, apply Lemma 3.1 to the finite set of elements represented by the generators, their inverses, and the prefixes of the relators and of the words of W , at a scale ε small against the lengths of those words; the value of a word of length ℓ at the model of the generators then lies within $2\ell\varepsilon$ of the model of the element it represents. If one of the challenged words is trivial, use the first answer; otherwise the two strict inequalities hold with room to spare. Conversely, choose a word representative r_g of each element $g \in G_P$, with r_1 the empty word. For a finite set $F \ni 1$ of elements put $W_F = \{r_g r_h^{-1} : g, h \in F, g \neq h\}$; every word in W_F is nontrivial in G_P , so the challenge (W_F, k) has a matrix answer U , and $V(g) = r_g(U)$ satisfies $\|V(g) - V(h)\| > 1/3$ for $g \neq h$ in F . For $g, h, gh \in F$ the word $r_g r_h r_{gh}^{-1}$ is trivial in G_P ; fix an expression of it as a product of conjugates of relators and their inverses, and let A_F bound the number of factors over all such triples. Then $\|V(g)V(h) - V(gh)\| \leq A_F/(k+1)$, which tends to zero as $k \rightarrow \infty$. This is the MF condition.

The matrix answers are coded as finite data. A *void answer* is a word $w \in W$ and a finite search witness exhibiting w as a product of conjugates of the relators and their inverses. A *matrix answer* is a positive dimension, Gaussian-rational matrices for the generators, one upper-norm certificate for each relator, and one lower-norm certificate for each word of W . The checker verifies exact unitarity; for upper bounds it uses the power and Frobenius estimates for X^*X , and for lower bounds a coded rational vector. So every field is finite data and the complete answer checker is primitive recursive. Every strict matrix answer in (1) has such a coded rational answer: the Cayley transform $(1 + ia)(1 - ia)^{-1}$ of a Gaussian-rational Hermitian matrix a is a Gaussian-rational unitary, and these unitaries are dense in $\mathcal{U}(d)$; soundness of the norm certificates gives the converse.

Fix total computable decoders $n \mapsto (W_n, k_n)$ for challenges and $c \mapsto a_c$ for answers, both surjective, and let $C(P, n, c)$ mean that the answer checker accepts a_c for (W_n, k_n) . Then C is decidable and (1) becomes

$$P \in \text{MF}_{\text{fp}} \iff \forall n \exists c C(P, n, c).$$

So $\text{MF}_{\text{fp}} \in \Pi_2^0$, and taking complements gives $\text{NONMF}_{\text{fp}} \in \Sigma_2^0$. $\square$

4. HNN EXTENSIONS WITH A CORONA CONJUGATOR

If the edge isomorphism of an HNN extension is implemented by a unitary of a norm matrix corona, then the HNN extension is MF. For $\theta = \text{id}$ this is close to Shulman's theorem on central HNN extensions [Sh]; the version below assumes a tracial state vanishing off the identity and allows a nontrivial edge isomorphism. The corona unitary implements the edge isomorphism, and the trace and the left regular representation show that the group embeds in the resulting algebra.

A countable group G admits a *tracial MF realization* (A, ρ, τ) if A is a separable unital MF C^* -algebra, $\rho: G \rightarrow \mathcal{U}(A)$ is a homomorphism, and τ is a tracial state on A with $\tau(\rho(g)) = 0$ for every $g \neq 1$. Since $\tau(1) = 1$, such a

group embeds in $\mathcal{U}(A)$, so it is MF. Only the vanishing of τ off the identity is used below, and the constructions below produce tracial states. To embed an arbitrary norm matrix corona $\mathcal{Q}_{\mathbf{d}}$ in $\prod_k M_k(\mathbb{C})/\oplus_k M_k(\mathbb{C})$, choose a strictly increasing sequence (m_n) with $m_n \geq d_n$, embed $M_{d_n}(\mathbb{C})$ as a corner of $M_{m_n}(\mathbb{C})$, and put zero in the unused coordinates. The resulting map on reduced products is an isometric $*$ -homomorphism, so the MF algebras of Section 2 are the separable C^* -algebras that embed in the latter corona, as in [BK, Sh].

Lemma 4.1 (residually finite groups). *Every countable residually finite group admits a tracial MF realization.*

Proof. Choose finite-index normal subgroups $N_1 \geq N_2 \geq \dots$ with trivial intersection, and let λ_k be the left regular representation of G/N_k , of dimension d_k . In the corona $\mathcal{Q}_{\mathbf{d}}$ the map $\rho(g) = [(\lambda_k(g))_k]$ is a homomorphism, and $\tau([(x_k)_k]) = \lim_{\omega} \text{tr}_{d_k}(x_k)$, for a free ultrafilter ω , is a tracial state with $\tau(\rho(g)) = 0$ for $g \neq 1$, since $g \notin N_k$ for all large k . The C^* -algebra generated by $\rho(G)$ and the unit of the corona is separable and unital, and the triple with ρ and τ is the required realization. $\square$

Let G be a countable group, let $S \leq G$ be a subgroup, and let $\theta: S \rightarrow G$ be an injective homomorphism. The HNN extension

$$(2) \quad R = \langle G, t \mid tst^{-1} = \theta(s) \ (s \in S) \rangle$$

contains G by Britton's lemma [Bri].

Theorem 4.2 (HNN permanence with a corona conjugator). *In the situation of (2), suppose that G admits a tracial MF realization (A, ρ, τ) , and that for some norm matrix corona $\mathcal{Q}$ there are an injective $*$ -homomorphism $\iota: A \rightarrow \mathcal{Q}$ and a unitary $W \in \mathcal{Q}$ with*

$$W \iota \rho(s) W^* = \iota \rho(\theta(s)) \quad (s \in S).$$

Then R admits a tracial MF realization. In particular, R is MF.

Proof. Write $D = C^*(\iota \rho(G)) \subseteq \mathcal{Q}$, a separable C^* -algebra with unit $p = \iota(1)$, and let $B_0 = C^*(\iota \rho(S))$ and $B_1 = C^*(\iota \rho(\theta(S)))$, both containing p . Conjugation by W restricts to a $*$ -isomorphism $\Theta: B_0 \rightarrow B_1$ carrying $\iota \rho(s)$ to $\iota \rho(\theta(s))$. Let U be the quotient of the full unital free product $D * C(\mathbb{T})$ by the closed ideal generated by the elements $ubu^* - \Theta(b)$, $b \in B_0$, where u is the canonical unitary generator of $C(\mathbb{T})$. We write d and u for the images of $d \in D$ and u in U . By construction, U has the following universal property: whenever $\pi: D \rightarrow E$ is a unital $*$ -homomorphism into a unital C^* -algebra and $v \in \mathcal{U}(E)$ satisfies $v\pi(b)v^* = \pi(\Theta(b))$ for $b \in B_0$, there is a unique $*$ -homomorphism $U \rightarrow E$ extending π and sending u to v . The algebra U embeds in an amalgamated free product, which Shulman's criterion shows to be MF, and the copy of R inside U carries a tracial state.

Step 1: U embeds in an amalgamated free product. Let $C = B_0 \oplus B_1$, $A_1 = M_2(D)$, and $A_2 = M_2(B_0)$, with the unital inclusions

$$\iota_A(c_0, c_1) = \begin{pmatrix} c_0 & 0 \\ 0 & c_1 \end{pmatrix} \in A_1, \quad \iota_B(c_0, c_1) = \begin{pmatrix} c_0 & 0 \\ 0 & \Theta^{-1}(c_1) \end{pmatrix} \in A_2,$$

and let $P = A_1 *_C A_2$ be the full amalgamated free product, in which A_1 and A_2 embed [Sh, Section 2.1]. Ueda proved that the universal HNN extension is a full corner of this amalgamated free product [Ue, Proposition 2.4]: for the projection $e = \iota_A(1_D, 0) = \iota_B(1_D, 0)$ there is an isomorphism of U onto ePe carrying $d \in D$ to $\text{diag}(d, 0)$ and the unitary u to the partial isometry $e_{12}f_{21}$ built from the matrix units e_{ij} of A_1 and f_{ij} of A_2 . In particular U embeds in P .

Step 2: P is MF. By Shulman's amalgamated-free-product criterion [Sh, Theorem 16], the full amalgamated free product $A_1 *_C A_2$ of separable C^* -algebras is MF as soon as there are injective $*$ -homomorphisms φ_A and φ_B of A_1 and A_2 into one norm matrix corona with $\varphi_A \circ \iota_A = \varphi_B \circ \iota_B$. Identify $M_2(\mathcal{Q})$ with the norm matrix corona of the doubled dimension sequence, and let $\varphi_A: M_2(D) \rightarrow M_2(\mathcal{Q})$ be the entrywise inclusion and $\varphi_B: M_2(B_0) \rightarrow M_2(\mathcal{Q})$ the entrywise inclusion followed by conjugation by $\text{diag}(1, W)$. Both are injective, and on C ,

$$\varphi_B(\iota_B(c_0, c_1)) = \begin{pmatrix} c_0 & 0 \\ 0 & W\Theta^{-1}(c_1)W^* \end{pmatrix} = \begin{pmatrix} c_0 & 0 \\ 0 & c_1 \end{pmatrix} = \varphi_A(\iota_A(c_0, c_1)),$$

because Θ is conjugation by W . So P is MF, as is its C^* -subalgebra $ePe \cong U$.

Step 3: the tracial state. The functional $T = \tau \circ \iota^{-1}$ is a tracial state on $\iota(A) \supseteq D$ with $T(\iota\rho(g)) = 0$ for $g \neq 1$. In the GNS representation of D for T , the vectors $\xi_g = \iota\rho(g)\xi_T$, $g \in G$, are orthonormal, because $\langle \xi_g, \xi_h \rangle = T(\iota\rho(h^{-1}g))$. Their closed span is invariant under D , and on it $\iota\rho(g)$ acts as the left regular representation $\lambda_G(g)$ of G . So there is a $*$ -homomorphism $\pi: D \rightarrow C_r^*(G)$ with $\pi(\iota\rho(g)) = \lambda_G(g)$. Since $G \leq R$, the space $\ell^2(R)$ is a direct sum of copies of $\ell^2(G)$ indexed by the right cosets of G , so $\lambda_R|_G$ is a multiple of λ_G and the assignment $\lambda_G(g) \mapsto \lambda_R(g)$ extends to a $*$ -homomorphism $C_r^*(G) \rightarrow C_r^*(R)$. Composing, we obtain $\sigma_0: D \rightarrow C_r^*(R)$ with $\sigma_0(\iota\rho(g)) = \lambda_R(g)$. The pair $(\sigma_0, \lambda_R(t))$ satisfies the covariance relation $\lambda_R(t)\sigma_0(b)\lambda_R(t)^* = \sigma_0(\Theta(b))$ for $b \in B_0$: on the generators $\iota\rho(s)$ both sides equal $\lambda_R(\theta(s))$, because $tst^{-1} = \theta(s)$ in R , and two $*$ -homomorphisms that agree on generators agree on B_0 . The universal property of U gives $\sigma: U \rightarrow C_r^*(R)$ with $\sigma(\iota\rho(g)) = \lambda_R(g)$ and $\sigma(u) = \lambda_R(t)$.

Define $j: R \rightarrow \mathcal{U}(U)$ by $j(g) = \iota\rho(g)$ for $g \in G$ and $j(t) = u$. The defining relations of (2) hold in U , since $u\iota\rho(s)u^* = \Theta(\iota\rho(s)) = \iota\rho(\theta(s))$, so j is a homomorphism, and $\sigma \circ j = \lambda_R$ is injective, so j is injective. Let $A' = C^*(j(R)) \subseteq U$, a separable unital MF C^* -algebra, and let $\tau' = \tau_R \circ \sigma|_{A'}$, where τ_R is the canonical trace $x \mapsto \langle x\delta_1, \delta_1 \rangle$ of $C_r^*(R)$. Then τ' is a tracial state on A' with $\tau'(j(r)) = \tau_R(\lambda_R(r)) = 0$ for $r \neq 1$. So (A', j, τ') is a tracial MF realization of R . $\square$

Injectivity of j comes from $\sigma \circ j = \lambda_R$; the covariant pair $(\iota\rho, W)$ need not be faithful on R , and in Section 5 it is not.

Corollary 4.3 (central HNN extensions). *If G admits a tracial MF realization and $S \leq G$ is any subgroup, then $\langle G, t \mid [t, s] = 1 \ (s \in S) \rangle$ admits a tracial MF realization.*

Proof. Apply Theorem 4.2 with θ the inclusion of S , with ι any injective $*$ -homomorphism of A into a norm matrix corona, and with $W = 1$. $\square$

The positive branch of the reduction below produces tracial MF realizations inside norm reduced products. For C^* -algebras B_n , $n \geq 1$, write $\prod_n B_n / \bigoplus_n B_n$ for the quotient of the bounded sequences by the sequences with $\|x_n\| \rightarrow 0$; its norm is $\limsup_n \|x_n\|$.

Lemma 4.4 (reduced products of MF algebras). *If every B_n is MF, then every separable C^* -subalgebra of $\prod_n B_n / \bigoplus_n B_n$ is MF. Moreover, $M_k(B)$ is MF whenever B is.*

Proof. The first assertion follows by diagonalization from the matrix-corona characterization of MF algebras [BK], and Korchagin uses it in this form [Kor, Proposition 6]. The second holds because M_k of a norm matrix corona is the norm matrix corona of the dimension sequence multiplied by k . $\square$

Lemma 4.5 (tensor synchronization). *Let Γ be a countable group with a tracial MF realization (A_1, ρ_1, τ_1) , let Q be a countable group with homomorphisms $\beta_n: Q \rightarrow B_n$ to finite groups such that every $q \neq 1$ satisfies $\beta_n(q) \neq 1$ for all large n , let $S \leq \Gamma$, and let $\tau: S \rightarrow Q$ be a homomorphism. Suppose that for every n there is a homomorphism $\lambda_n: \Gamma \rightarrow G_n$ to a finite group with*

$$\ker(\lambda_n|_S) \leq \ker(\beta_n \circ \tau).$$

Then there are a separable unital MF C^ -algebra A , a homomorphism $V: \Gamma \times Q \rightarrow \mathcal{U}(A)$, a tracial state T on A with $T(V(g, q)) = 0$ for $(g, q) \neq (1, 1)$, and a unitary $W \in A$ with $WV(s, 1)W^* = V(s, \tau(s))$ for all $s \in S$. Consequently*

$$\langle \Gamma \times Q, u \mid u(s, 1)u^{-1} = (s, \tau(s)) \ (s \in S) \rangle$$

admits a tracial MF realization, and in particular is MF.

Proof. Let E_n be the image of $(\lambda_n, \beta_n): \Gamma \times Q \rightarrow G_n \times B_n$, a finite group of order k_n , and let L_n be its left regular representation on $\ell^2(E_n)$. Put $B'_n = A_1 \otimes M_{k_n}(\mathbb{C})$, which is MF by Lemma 4.4, let $E = \prod_n B'_n / \bigoplus_n B'_n$, and define

$$V(g, q) = [(\rho_1(g) \otimes L_n(\lambda_n(g), \beta_n(q)))_n], \quad T([(x_n)_n]) = \lim_{\omega} (\tau_1 \otimes \text{tr}_{k_n})(x_n),$$

for a free ultrafilter ω . Each coordinate of V is a homomorphism into a unitary group, so V is a homomorphism, and T is a tracial state on E because $|(\tau_1 \otimes \text{tr}_{k_n})(x_n)| \leq \|x_n\|$. If $g \neq 1$ then $\tau_1(\rho_1(g)) = 0$, so $T(V(g, q)) = 0$; if

$g = 1$ and $q \neq 1$ then $\beta_n(q) \neq 1$ for all large n , the regular representation of a finite group has trace zero off the identity, and again $T(V(1, q)) = 0$.

The two homomorphisms $s \mapsto (\lambda_n(s), 1)$ and $s \mapsto (\lambda_n(s), \beta_n \tau(s))$ from S to E_n have the same kernel, namely $\ker(\lambda_n|_S)$, by the hypothesis. So their images are isomorphic subgroups of E_n of the same order, isomorphic via $(\lambda_n(s), 1) \mapsto (\lambda_n(s), \beta_n \tau(s))$. The restriction of L_n to a subgroup of E_n is unitarily equivalent to the direct sum of as many copies of the left regular representation of that subgroup as it has right cosets, so there is a unitary $W_n \in M_{k_n}(\mathbb{C})$ with $W_n L_n(\lambda_n(s), 1) W_n^* = L_n(\lambda_n(s), \beta_n \tau(s))$ for all $s \in S$. Put $W = [(1 \otimes W_n)_n] \in E$; then $WV(s, 1)W^* = V(s, \tau(s))$ for $s \in S$. Let A be the C^* -algebra generated by $V(\Gamma \times Q)$ and W ; it is separable and unital, and MF by Lemma 4.4. The triple $(A, V, T|_A)$ is a tracial MF realization of $\Gamma \times Q$. For the last assertion, apply Theorem 4.2 to this realization: for an injective $*$ -homomorphism ι of A into a norm matrix corona with $p = \iota(1)$, the element $\widetilde{W} = \iota(W) + (1 - p)$ is a unitary of the full corona and implements the required covariance on $\iota(A)$. $\square$

5. THE REDUCTION

Write W_e for the domain of the e -th partial computable function and

$$\text{INF} = \{e : W_e \text{ is infinite}\}, \quad \text{FIN} = \{e : W_e \text{ is finite}\}.$$

The set INF is Π_2^0 -complete and FIN is Σ_2^0 -complete [So, Chapter IV]. This section constructs a computable map $e \mapsto \widehat{R}_e$ into finite presentations such that the presented group contains the non-MF group E when $e \in \text{FIN}$ and satisfies the hypotheses of Theorem 4.2 when $e \in \text{INF}$. Table 1 lists the objects of the construction.

The seed.

Lemma 5.1 (the seed). *There is a fixed finite presentation P_- whose group $E = G_{P_-}$ is not MF.*

Proof. Theorem C of [NMF] gives a finitely presented torsion-free group with no nontrivial homomorphism to an MF group; such a group is not MF unless it is trivial, and it is not. Fix one finite presentation code P_- for this group. $\square$

The switch.

Lemma 5.2 (the group C_e). *There is an algorithm that computes from e a recursive presentation, on a computable set of generators, of a group C_e such that C_e is trivial if $e \in \text{INF}$ and $C_e \cong E$ if $e \in \text{FIN}$.*

Proof. Write $P_- = \langle X \mid R \rangle$ with X finite. The generators of C_e are the symbols $x^{(k)}$ for $x \in X$ and $k \in \mathbb{N}$, one copy of X for each layer k . Run the e -th program on all inputs by dovetailing, and let m_s be the number of elements of W_e found by stage s , with $m_{-1} = 0$. The relators are enumerated in stages. At stage s , enumerate the relators R written in the layer s ; if

E	the finitely presented non-MF group of [NMF, Theorem C]
C_e	recursively presented; trivial for $e \in \text{INF}$, isomorphic to E for $e \in \text{FIN}$
$Q_e = F/N_e$	three-generated recursively presented group containing C_e ; here $F = F(x, y, t)$
$Q_+ = F/N_+$	the value of Q_e for trivial C_e ; embeds in $F(x_1, y) \times F(x_2, t)$
$\Lambda_e; M_e$	finitely presented Higman host of Q_e ; the Mihailova subgroup detecting equality in Λ_e
$K_e; L_e$	a product of six free groups; its finitely generated subgroup with $i(F) \cap L_e = i(N_e)$
$\Gamma_e; S_e$	the central HNN extension $\langle K_e, v \mid [v, \ell] = 1 \ (\ell \in L_e) \rangle$; $S_e = \langle i(F), v i(F) v^{-1} \rangle \cong F *_{N_e} F$
$\alpha_e: S_e \rightarrow Q_e$	q_e on $i(F)$, trivial on $v i(F) v^{-1}$
$\hat{R}_e$	the finite presentation of $\langle \Gamma_e \times Q_e, u \mid u(s, 1)u^{-1} = (s, \alpha_e(s)) \ (s \in S_e) \rangle$

Table 1: The objects of the construction, in order of appearance.

$m_s > m_{s-1}$, enumerate $x^{(k)} = 1$ for all $x \in X$ and all $k \leq s$; and otherwise enumerate $x^{(s)} = x^{(s+1)}$ for all $x \in X$. The enumerator is computable in P_- and e .

If $e \in \text{INF}$, then for every k some stage $s \geq k$ has $m_s > m_{s-1}$, so every generator is killed and C_e is trivial. If $e \in \text{FIN}$, let s_* be the last stage with $m_s > m_{s-1}$, or $s_* = -1$ if there is none. The layers $k \leq s_*$ are killed; the stage s_* identifies nothing, so layer s_* is not identified with layer $s_* + 1$; and every stage $s > s_*$ identifies layer s with layer $s + 1$. So the layers $k > s_*$ are identified with one another, and $x^{(k)} \mapsto x$ for $k > s_*$, $x^{(k)} \mapsto 1$ for $k \leq s_*$, is an isomorphism $C_e \rightarrow E$ with inverse $x \mapsto x^{(s_*+1)}$. $\square$

A three-generator embedding. Let $F = F(x, y, t)$ be the free group on x, y, t , and let $y_i = x^i y x^{-i}$ for $i \in \mathbb{Z}$. For a countable group C with a generating sequence $(c_i)_{i \geq 1}$, put $c_i = 1$ for $i \leq 0$ and

$$(3) \quad B(C) = \langle C * F(x, y), t \mid t y_i t^{-1} = c_i y_i \ (i \in \mathbb{Z}) \rangle.$$

Let $Q_+ = B(1) = \langle x, y, t \mid [t, y_i] = 1 \ (i \in \mathbb{Z}) \rangle$, let $q_+: F \rightarrow Q_+$ be the quotient map, and let $N_+ = \ker q_+$.

Lemma 5.3 (the three-generator embedding).

- (1) $B(C)$ is an HNN extension of $C * F(x, y)$, so C embeds in $B(C)$; it is generated by x, y, t ; and a recursive presentation of $B(C)$ on x, y, t is computable from a recursive presentation of C . We write $B(C) = F/N_C$ and $q_C: F \rightarrow B(C)$ for the quotient map.
- (2) Killing C induces a surjection $B(C) \rightarrow Q_+$; so $N_C \leq N_+$, with equality when C is trivial.

- (3) The assignment $x \mapsto (x_1, x_2)$, $y \mapsto (y, 1)$, $t \mapsto (1, t)$ defines an injective homomorphism $j: Q_+ \rightarrow P = F(x_1, y) \times F(x_2, t)$. In particular, Q_+ is residually finite.

Proof. (1) The elements y_i , $i \in \mathbb{Z}$, freely generate the normal closure of y in $F(x, y)$ [LS, Chapter I], and the retraction $C * F(x, y) \rightarrow F(x, y)$ killing C sends $c_i y_i$ to y_i , so the $c_i y_i$ freely generate a free subgroup and $y_i \mapsto c_i y_i$ is an isomorphism between two free subgroups of the base. So (3) is an HNN extension, and the base, and with it C , embeds by Britton's lemma. The relation $c_i = t y_i t^{-1} y_i^{-1}$ shows that x, y, t generate $B(C)$. If $C = \langle c_1, c_2, \dots \mid R_j(c_1, c_2, \dots) = 1 \ (j \geq 1) \rangle$, then eliminating the c_i with $i \geq 1$ by this relation and keeping the relations of (3) with $i \leq 0$ gives

$$B(C) = \langle x, y, t \mid R_j(t y_1 t^{-1} y_1^{-1}, t y_2 t^{-1} y_2^{-1}, \dots) = 1 \ (j \geq 1), [t, y_i] = 1 \ (i \leq 0) \rangle,$$

a recursive presentation on x, y, t , uniformly in the presentation of C .

(2) Killing every c_i sends the relators of (3) to the relators $[t, y_i]$ of Q_+ and is the identity on x, y, t .

(3) The relators $[t, y_i]$ are sent to $[(1, t), (x_1^i y x_1^{-i}, 1)] = 1$, so j is a homomorphism. The normal closure N_0 of $\{y, t\}$ in Q_+ is $F(y_n : n \in \mathbb{Z}) \times F(t_m : m \in \mathbb{Z})$ with $y_n = x^n y x^{-n}$ and $t_m = x^m t x^{-m}$, by the Reidemeister–Schreier method over the transversal $\{x^n\}$ [LS, Chapter II]; j carries these bases to free bases of the normal closures of y in $F(x_1, y)$ and of t in $F(x_2, t)$, and the exponent sum of x_1 in $j(x^k h)$, $h \in N_0$, is k . So j is injective [Lean]. Free groups are residually finite, and residual finiteness passes to direct products and subgroups. $\square$

For $e \in \mathbb{N}$ let $Q_e = B(C_e) = F/N_e$ with C_e the group of Lemma 5.2, and write $q_e: F \rightarrow Q_e$. By Lemmas 5.2 and 5.3, a recursive presentation of Q_e on x, y, t is computable from e ; $N_e \leq N_+$; if $e \in \text{INF}$ then $Q_e = Q_+$, $N_e = N_+$ and $q_e = q_+$; and if $e \in \text{FIN}$ then $E \leq C_e \leq Q_e$.

Detecting N_e inside a finitely presented group.

Lemma 5.4 (Higman embedding and the Mihailova subgroup). *There is an algorithm that computes from e a finite presentation $\langle X_e \mid R_e \rangle$ of a group Λ_e , words $w_x, w_y, w_t \in F(X_e)$, and a finite generating set of a subgroup $M_e \leq F(X_e) \times F(X_e)$, such that $x \mapsto w_x$, $y \mapsto w_y$, $t \mapsto w_t$ induces an embedding $Q_e \rightarrow \Lambda_e$, and $M_e = \{(u, v) : u = v \text{ in } \Lambda_e\}$. Consequently, with*

$$K_e^0 = F \times F(X_e) \times F(X_e), \quad i_e(f) = (f, w_f, 1), \quad L_e^0 = F \times M_e,$$

where w_f is the image of f under $x \mapsto w_x$, $y \mapsto w_y$, $t \mapsto w_t$, the map $i_e: F \rightarrow K_e^0$ is injective and $i_e(f) \in L_e^0$ if and only if $f \in N_e$.

Proof. Higman's embedding theorem [Hig] embeds every finitely generated recursively presented group in a finitely presented group, and Mikaelian [Mik] gives an explicit algorithm producing the finite presentation and the embedding words from the recursive presentation. Apply it to the presentation of Q_e . Let M_e be the subgroup of $F(X_e) \times F(X_e)$ generated by the pairs (z, z) ,

$z \in X_e$, and $(r, 1)$, $r \in R_e$; then $M_e = \{(u, v) : u = v \text{ in } \Lambda_e\}$ by Mihailova's lemma [Mih]. The first coordinate makes i_e injective, and $(f, w_f, 1) \in F \times M_e$ if and only if $w_f = 1$ in Λ_e , if and only if $q_e(f) = 1$, if and only if $f \in N_e$. $\square$

The two HNN extensions. Let $P = F(x_1, y) \times F(x_2, t)$ and $j: Q_+ \rightarrow P$ be as in Lemma 5.3, and put

$$K^g = F \times P, \quad L^g = \{(f, jq_+(f)) : f \in F\},$$

$$K_e = K_e^0 \times K^g, \quad L_e = L_e^0 \times L^g, \quad i: F \rightarrow K_e, \quad i(f) = (i_e(f), (f, 1)).$$

Let

$$(4) \quad \Gamma_e = \langle K_e, v \mid [v, \ell] = 1 \ (\ell \in L_e) \rangle, \quad S_e = \langle i(F), v i(F) v^{-1} \rangle \leq \Gamma_e.$$

Lemma 5.5 (the first HNN extension).

- (1) K_e is a direct product of free groups of finite rank, so finitely presented and residually finite; L_e is finitely generated; i is injective; and $i(F) \cap L_e = i(N_e)$.
- (2) Γ_e is finitely presented, and a finite presentation is computable from e . The natural map from the amalgamated free product $F *_{N_e} F$, in which $n \in N_e$ in the first copy is identified with n in the second copy, onto S_e , sending the first copy to $i(F)$ and the second to $v i(F) v^{-1}$, is an isomorphism.
- (3) There is a homomorphism $\alpha_e: S_e \rightarrow Q_e$ with $\alpha_e(i(f)) = q_e(f)$ and $\alpha_e(v i(f) v^{-1}) = 1$ for $f \in F$.

Proof. (1) $K_e = F \times F(X_e) \times F(X_e) \times F \times P$ is a product of six free groups of finite rank, so it is finitely presented and residually finite. The subgroup L_e is generated by the basis of the first factor, the generators of M_e from Lemma 5.4, and the elements $(a, jq_+(a))$ for $a \in \{x, y, t\}$. Injectivity of i comes from the first coordinate. An element $i(f)$ lies in L_e if and only if $i_e(f) \in L_e^0$ and $(f, 1) \in L^g$; the first holds if and only if $f \in N_e$ by Lemma 5.4, and the second if and only if $jq_+(f) = 1$, that is, $f \in N_+$. Since $N_e \leq N_+$, the intersection is $i(N_e)$.

(2) The presentation (4) is finite once $[v, \ell] = 1$ is imposed only for the finitely many generators ℓ of L_e , and it is computable from e because K_e , L_e and i are. It is an HNN extension of K_e with both edge groups equal to L_e and the identity as edge isomorphism. The map from $F *_{N_e} F$ to S_e is well defined because $i(n) \in L_e$ commutes with v for $n \in N_e$. A reduced word of the amalgamated free product alternates between elements of the two copies of F lying outside N_e ; its image is a word $i(f_0) v i(f_1) v^{-1} i(f_2) \cdots$ in which every letter between consecutive occurrences of $v^{\pm 1}$ is some $i(f_k)$ with $f_k \notin N_e$, so $i(f_k) \notin L_e$ by (1). Such a word contains no pinch $v^{\pm 1} \ell v^{\mp 1}$ with $\ell \in L_e$, so it is nontrivial in Γ_e by Britton's lemma.

(3) Define α_e on $F *_{N_e} F$ by q_e on the first copy and by the trivial map on the second; both kill N_e , so the definition is consistent on the amalgamated subgroup, and (2) identifies $F *_{N_e} F$ with S_e . $\square$

The second HNN extension is

$$(5) \quad R_e = \langle \Gamma_e \times Q_e, u \mid u(s, 1)u^{-1} = (s, \alpha_e(s)) \ (s \in S_e) \rangle,$$

whose edge maps $s \mapsto (s, 1)$ and $s \mapsto (s, \alpha_e(s))$ are injective homomorphisms of S_e into $\Gamma_e \times Q_e$; so $\Gamma_e \times Q_e$ embeds in R_e by Britton's lemma. The presentation (5) lists the recursively enumerable relators of Q_e . Its finite replacement is

$$(6) \quad \begin{aligned} \hat{R}_e = \langle \Gamma_e \times F, u \mid & u(i(a), 1)u^{-1} = (i(a), a), \\ & u(vi(a)v^{-1}, 1)u^{-1} = (vi(a)v^{-1}, 1) \ (a \in \{x, y, t\}) \rangle, \end{aligned}$$

where $\Gamma_e \times F$ is equipped with the finite presentation of Lemma 5.5(2), the basis x, y, t of F , and the commutation relators between the two factors.

Lemma 5.6 (the finite presentation). *The groups $\hat{R}_e$ and R_e are isomorphic by the map that is the identity on Γ_e and on u and sends F onto Q_e by q_e . A code of the finite presentation (6) is computable from e .*

Proof. Both maps $f \mapsto u(i(f), 1)u^{-1}$ and $f \mapsto (i(f), f)$ are homomorphisms $F \rightarrow \hat{R}_e$, and they agree on the basis, so $u(i(f), 1)u^{-1} = (i(f), f)$ for all $f \in F$; likewise $u(vi(f)v^{-1}, 1)u^{-1} = (vi(f)v^{-1}, 1)$ for all f . If $n \in N_e$, then $i(n) \in L_e$ by Lemma 5.5(1), so $vi(n)v^{-1} = i(n)$ in Γ_e , and comparing the two relations gives $(i(n), n) = (i(n), 1)$, that is, $(1, n) = 1$ in $\hat{R}_e$. So the factor F of $\hat{R}_e$ factors through $Q_e = F/N_e$, and the relations of (6) become $u(s, 1)u^{-1} = (s, \alpha_e(s))$ for s in the generating set $\{i(a), vi(a)v^{-1} : a \in \{x, y, t\}\}$ of S_e ; both sides define homomorphisms on S_e , so the relations hold for all $s \in S_e$. This defines a homomorphism $R_e \rightarrow \hat{R}_e$; conversely, the relations of (6) hold in R_e after $F \rightarrow Q_e$, because $\alpha_e(i(a)) = q_e(a)$ and $\alpha_e(vi(a)v^{-1}) = 1$. The two homomorphisms are mutually inverse on generators. Computability of the code follows from Lemmas 5.4 and 5.5. $\square$

The two branches.

Lemma 5.7 (the case $e \in \text{FIN}$). *If $e \in \text{FIN}$, then E embeds in $\hat{R}_e$, and $\hat{R}_e$ is not MF.*

Proof. For $e \in \text{FIN}$ we have $E \leq C_e \leq Q_e \leq \Gamma_e \times Q_e \leq R_e \cong \hat{R}_e$ by Lemmas 5.2, 5.3(1), and 5.6, and Britton's lemma for (5). Subgroups of MF groups are MF, and E is not MF by Lemma 5.1. $\square$

Lemma 5.8 (the case $e \in \text{INF}$). *If $e \in \text{INF}$, then $\hat{R}_e$ admits a tracial MF realization; in particular, it is MF.*

Proof. Let $e \in \text{INF}$, so that $Q_e = Q_+$, $N_e = N_+$, $q_e = q_+$, and $\alpha_e(i(f)) = q_+(f)$, $\alpha_e(vi(f)v^{-1}) = 1$. By Lemma 5.6 it suffices to show that R_e admits a tracial MF realization, and by Lemma 4.5 applied to $\Gamma = \Gamma_e$, $Q = Q_+$, $S = S_e$ and $\tau = \alpha_e$, it suffices to produce the data of that lemma.

The group K_e is residually finite by Lemma 5.5(1), so it admits a tracial MF realization by Lemma 4.1; and Γ_e , the central HNN extension of K_e over L_e , admits one by Corollary 4.3.

The group $P = F(x_1, y) \times F(x_2, t)$ is residually finite; choose finite-index normal subgroups $P_1 \geq P_2 \geq \dots$ of P with trivial intersection, let $r_n: P \rightarrow C_n = P/P_n$, and put $\beta_n = r_n \circ j: Q_+ \rightarrow C_n$ with j the embedding of Lemma 5.3(3). Since j is injective and the P_n decrease to the trivial group, every $h \neq 1$ has $\beta_n(h) \neq 1$ for all large n .

Let $G_n = (C_n \times C_n) \rtimes \langle \sigma \rangle$, where σ of order two exchanges the coordinates. Define λ_n on $K_e = K_e^0 \times (F \times P)$ by killing K_e^0 and sending (f, p) to $(r_n j q_+(f), r_n(p))$, and put $\lambda_n(v) = \sigma$. This respects the relations of (4): the factor L_e^0 of L_e lies in K_e^0 and is killed, and an element $(f, j q_+(f))$ of L_e^g is sent to $(r_n j q_+(f), r_n j q_+(f))$, which lies on the diagonal and commutes with σ . So $\lambda_n: \Gamma_e \rightarrow G_n$ is a homomorphism, and on the generators of S_e ,

$$\lambda_n(i(f)) = (r_n j q_+(f), 1), \quad \lambda_n(v i(f) v^{-1}) = (1, r_n j q_+(f)).$$

Let $\pi_0, \pi_1: S_e \rightarrow Q_+$ be the homomorphisms that are q_+ on one copy of F in $S_e \cong F *_{N_+} F$ and trivial on the other, as in Lemma 5.5(3); thus $\pi_0 = \alpha_e$. The displayed formulas show that $\lambda_n|_{S_e} = (r_n j \pi_0, r_n j \pi_1)$, both sides being homomorphisms that agree on generators. So

$$\ker(\lambda_n|_{S_e}) = \ker(r_n j \pi_0) \cap \ker(r_n j \pi_1) \leq \ker(\beta_n \circ \alpha_e),$$

which is the hypothesis of Lemma 4.5. $\square$

The second coordinate of G_n is needed: with $\lambda_n(v) = 1$ the two copies of F in S_e would have the same image, and the kernel condition would fail on $i(f) v i(f)^{-1} v^{-1}$ for $f \notin N_+$.

Proof of Theorem A. By Lemma 5.6 the map $e \mapsto \widehat{R}_e$ is computable, by Lemma 5.8 the presented group is MF for $e \in \text{INF}$, and by Lemma 5.7 it is not MF for $e \in \text{FIN}$. So INF reduces to MF_{fp} and FIN reduces to NONMF_{fp} under computable many-one reductions. Since INF is Π_2^0 -complete and FIN is Σ_2^0 -complete, and since $\text{MF}_{\text{fp}} \in \Pi_2^0$ and $\text{NONMF}_{\text{fp}} \in \Sigma_2^0$ by Proposition 3.2, the two sets are complete at their levels. A Π_2^0 -complete set is not Σ_2^0 and a Σ_2^0 -complete set is not Π_2^0 , so neither set is recursively enumerable, and in particular neither is decidable. $\square$

Remark 5.9. The construction uses two properties of E : it is finitely presented, and it embeds in no MF group. So it applies to any group property that is closed under subgroups, given a finitely presented group without the property and an analogue of Lemma 4.5 for the twisted HNN extensions; the open problems below state which input is missing for soficity and for hyperlinearity.

Open problems. We do not know whether the models of $\widehat{R}_e$ can be chosen to converge to $C_r^*(\widehat{R}_e)$ in operator norm, so that the reduced C^* -algebra itself is MF; the argument produces a tracial state on an abstract MF algebra. We do not know whether $\widehat{R}_e$ is hyperlinear for $e \in \text{INF}$; placing

hyperlinearity recognition at the same level would also require a finitely presented non-hyperlinear group, and none is known. We do not know whether an analogous family of finitely presented groups exists for soficity; such a construction would require a permutation-model replacement for Lemma 4.5. Finally, we do not know the complexity of MF recognition under the strong convergence convention of [GKM, Sch]: the norms $\|\lambda(w) - 1\|$ in the left regular representation equal 0 for $w = 1$ and are at least $\sqrt{2}$ for $w \neq 1$, since $\|(\lambda(w) - 1)\delta_1\| = \|\delta_w - \delta_1\|$, so an algorithm approximating them uniformly in the finite presentation would solve the word problem, and the certificate argument of Proposition 3.2 does not transfer.

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